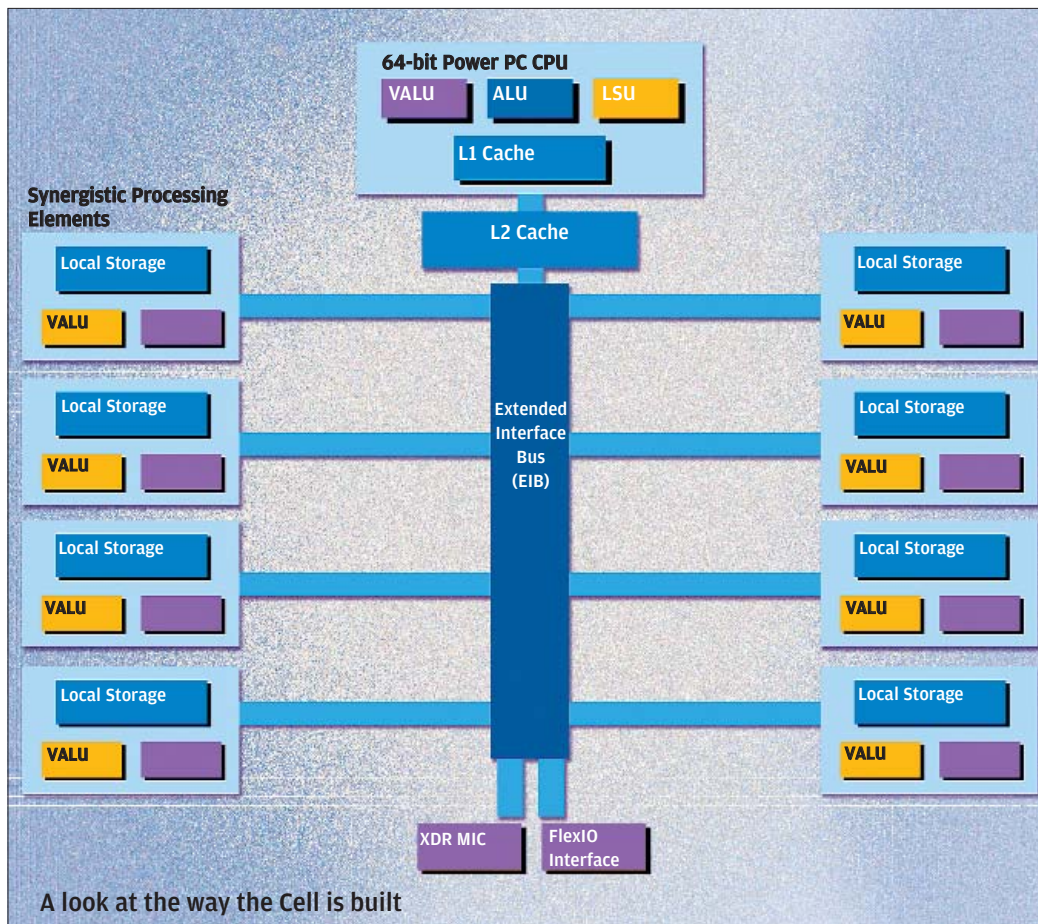




The Package

The Cell consists of a single PowerPC Processing Element (PPE) and eight identical Synergistic Processing Elements (SPE). The PPE is a simpler version of IBM's previous PowerPC processors (more on this in a bit), and basically functions as the general-purpose core in this collection. The eight SPEs are specialised SIMD (Single Instruction, Multiple Data; see box *SIMD*) processors, designed specifically to make quick work of complex mathematical problems. With the SPEs, IBM has removed the cache, substituting it instead with 256 KB of local storage—the name has changed, but the purpose remains the same; but read on to find out why it's not really cache.

The Cell has been optimised to work with parallel workloads—and few things can be broken into as many

Wars." This was all very well for a while, but there was also the matter of executing more instructions in each cycle of that clock signal. Naturally, the way to do this is have instructions run simultaneously—the concept is called Instruction Level Parallelism (ILP), and goes on in your CPU right now. The processor collects instructions, sees which it can run in parallel, and then does so.

Now, your average computer program is a single-threaded application—put simply, churning out a single stream of instructions, oblivious to the fact that there is a processor under the hood that's trying to figure out which of those instructions it can run simultaneously to increase performance. The result? Every new generation of processors delivered only a 10 to 20 per cent boost in performance: quite distressing.

But things are changing. The GigaHertz wars are over, and everyone has realised that the only way to squeeze more performance is to have multiple program threads running at the same time—Thread Level Parallelism (TLP). The way to do this is to increase the number of processors doing the work—first with the powerful dual-processor workstations, and now today's dual-core, tomorrow's quad-core, and in five years, eighty-core processors (at least, that's what Intel tells us).

While Intel and AMD have chosen to have identical cores on their chips, IBM, as usual, decided to do things a little differently.

parallel processes as games. Rendering frames requires processing millions of pixels, and this task can be broken into as many parallel tasks as one wants. Ditto physics and AI calculations. The Cell's obvious leaning towards better physics and AI means game scenarios can get more crowded (the more objects you place in a scene, the more complex physics calculations become) without any loss in performance. In fact, the Cell is even capable of taking up the job of the GPU, though industry experts suggest that it won't be replacing the dedicated GPU just yet.

The Cell's main purpose in the PS3 is to ensure that there are no bottlenecks whatsoever, so obscene amounts of memory bandwidth are the order of the day. You no doubt remember Rambus—the people who came up with RDRAM—a technology that was undoubtedly superior to DDR-SDRAM, but never really took off. Cell sports Rambus' new XDR memory controller, which gives each processor core a memory bandwidth of 25.6 GBps—more than twice that of any PC processor. This also comes very close to the 32 GBps that GPUs get from their memory controllers, so the Cell won't bottleneck the GPU the way PC processors do.

But what's so special about these processors that make them so fast? The secret to Cell's performance is a...

...Return To Innocence

The Cell's cores are all *in-order* cores—a design that hasn't been around since the days of the old